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SOIL PHYSICS & HYDROLOGY NOTES

soilphysics: A suite of tools to fit and compare water retention curves

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Abstract

Water content plays a central role in soil physical dynamic processes. The relationship between water content and matric potential is key to understand many of these processes. However, there is not one but at least half a dozen parametric models that can properly describe that relationship, which might have different levels of complexity, varying from soil to soil. Thus, fitting some of those models is not always a trivial task. Moreover, few computational resources are currently available to do this; or else, they can be of complex use as the models get more complex. Here, we present a suite of functionalities of the R package *soilphysics* that can be used to fit and statistically compare five water retention curve models, based on uni- or bimodal pore size distribution, using different approaches, from command line to interactive and automatized tools in user-friendly interfaces. The interactive approaches can also serve as didactic mechanisms, as they can help the user to understand how the water retention curve model fits the experimental data. *soilphysics* is freely available from the Comprehensive R Archive Network (CRAN; https://CRAN.R-project.org/package=soilphysics). The functionalities presented here are available from versions 4.1 or later; some tools can also run online.

1 | INTRODUCTION

The unsaturated zone of agricultural soils is of particular interest for plant growth, microorganism, and nutrient dynamics. Soil physical processes such as drainage, retention, and aeration are highly dependent on the energy state for which water is retained in the soil's porous system. The pore size distribution (structure and texture dependent) in the soil governs the dynamics of water flux, being commonly characterized by the water retention curve—the relationship between the soil water content and matric potential (Du, 2020; Shukla, 2013).

Once an experimental relationship between water content and matric potential for a specific soil sample is obtained, that relationship needs to be well described for a given matric potential range through an equation, in order to retrieve further information on the hydraulic dynamics (Groenevelt & Grant, 2004). Due to the heterogeneity of pore size distribution across soils around the world, many equations to describe the water retention curve have been proposed (Assouline et al., 1998; Brooks & Corey, 1964; Dexter et al., 2008; Fayer & Simmons, 1995; Fredlund & Xing, 1994; Iden & Durner, 2014; Ippisch et al., 2006; Khlosi et al., 2006; Peters, 2013; Rossi & Nimmo, 1994; Satyanaga et al., 2017; van Genuchten, 1980; Vanderlinden et al., 2017; Wang et al., 2018; Webb, 2000; Zhai et al., 2019).

An appropriate soil water retention model is essential for providing realistic descriptions of soil water behavior over wet and dry ranges. The Brooks and Corey (1964) and van Genuchten (1980) models are the most usual equations for description of wet range, but these two models do not describe

Abbreviations: ML, maximum likelihood; NLS, nonlinear weighted least squares.

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well the water content–matric potential relationship at oven dryness range, because they assume the matric potential to be infinite when the soil water approaches the residual water content (Du, 2020). On the other hand, Groenevelt and Grant (2004) developed an equation to better describe the dry end of the water retention curve, in an attempt to solve the problem of the residual water content.

Usually, a more complex the pore size distributions requires a more complex equation (Dexter et al., 2008; Durner, 1994; Fredlund & Xing, 1994; Satyanaga et al., 2017). Because those models are nonlinear, the parameter estimation methods need a numerical solution (Dourado-Neto et al., 2000; Seki, 2007; van Genuchten et al., 1991). Despite recent computational developments, user-friendly and open access computational environments for fitting water retention curves are still scarce. In this technical note, we present the usual water retention models and a suite of functionalities of the R package *soilphysics* with its implemented procedures to fit and statistically compare five soil water retention models.

2 | SOIL WATER RETENTION MODELS

To avoid any confusion in the mathematical formulation, the matric potential (Ψ) will be treated in terms of water tension (h), given by $h = |\Psi|$, in hPa ($\sim$ cm) (unit also used in the software). In addition, the error term ε of the models will be omitted for simplicity, keeping only the θ–h functional form, where θ is the volumetric water content, given in $m^3\ m^{-3}$.

Five equations to describe soil water retention curves are currently available in *soilphysics*. They were implemented based on their wide applicability, frequent use, and capability to describe the pore size distribution and to characterize the dry range of the θ–h relationship. The curves will be sequentially described according to their publication chronology.

2.1 | Brooks and Corey model

Equation 1 in Brooks & Corey (1964) is one of the first parametric models to describe the θ–h relationship for soils with an unimodal pore size distribution:

$$\theta = \theta_s, \quad \text{for } h \le h_b \tag{1}$$

$$\theta = \theta_r + (\theta_s - \theta_r)\left(\frac{h_b}{h}\right)^{\lambda}, \quad \text{for } h > h_b$$

where θ_s ($m^3\ m^{-3}$) is water content at saturation, and θ_r ($m^3\ m^{-3}$) is the residual water content; h_b (hPa) and λ (dimensionless) are fitting parameters, known as the bubbling potential or air-entry potential and the pore size distribution parameter (Du, 2020; Shukla, 2013), respectively.

Core Ideas

- A computational tool for fitting water retention curves is presented.
- The tool is developed in shiny R environment.
- Five water retention curves are available.
- The interactive approaches can also serve as a didactic tool.
- This tool is freely available in the R package *soilphysics*.

2.2 | van Genuchten model

The most widely used water retention curve was developed by van Genuchten (1980), also for soils with an unimodal pore size distribution. Equation 2 can fit and describe well water retentions phenomena of most soils (Radcliffe & Šimůnek, 2018):

$$\theta = \theta_r + \frac{\theta_s - \theta_r}{\left[1 + (\alpha h)^n\right]^m} \tag{2}$$

where θ_s ($m^3\ m^{-3}$) is water content at the saturation, θ_r ($m^3\ m^{-3}$) is the residual water content, and α (hPa^{-1}), n, and m are fitting parameters (dimensionless), with α assuming units equivalent to those of h. According to van Genuchten's (1980), m can be reparametrized using the Mualem's (1976) restriction, $m = 1 - 1/n$.

2.3 | Durner model

Durner (1994) stated that errors in the distribution of pores via the water retention curve may arise from the model that is used to describe that phenomenon. The author observed that the pore size distribution of some soils can be better described by an Equation 3 that fits a bimodal pore system:

$$\theta = \theta_r + (\theta_s - \theta_r)\left\{\begin{matrix} w_1\left[1 + (\alpha_1 h)^{n_1}\right]^{\frac{1}{n_1}-1} \\ +w_2\left[1 + (\alpha_2 h)^{n_2}\right]^{\frac{1}{n_2}-1} \end{matrix}\right\} \tag{3}$$

Durner (1994) uses the van Genuchten's equation to formulate this model; therefore, θ_r, θ_s, α, and n are the parameters described in the section above, including the Mualem's (1976) restriction. The additional parameters are w_1 and w_2 (dimensionless, where the sum is equal to 1), which are weighting values associated with the contributions of structural and textural porosities (Asgarzadeh et al., 2014).

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2.4 | Groenevelt and Grant model

According to Groenevelt and Grant (2004), most fitting equations to water retention curves are anchored only at the saturated water content, θ_s, and because they are not also anchored at the dry end, this causes what they refer to a "problem of the residual water content, θ_r." So, they have proposed an Equation 4 to better describe the θ–*h* dry end:

$$\theta = k_1 \left[\exp\left(\frac{-k_0}{\mathrm{pF}_0^n}\right) - \exp\left(\frac{-k_0}{\mathrm{pF}^n}\right) \right] \tag{4}$$

Where k_1, k_0, and n are fitting parameters, $\mathrm{pF} = \log_{10}(h)$ (for h given in hPa), and pF_0 is the pF value when the soil water content becomes zero (anchor point of the curve). Koorevaar et al. (1983) found that $\theta = 0$ at $\mathrm{pF} = 7.0$ for a wide range of soils (from a dune-sand to a river-basin clay, and ranging in composition from calcareous to organic) (Groenevelt & Grant, 2004), whereas Groenevelt and Grant (2004) suggested $\mathrm{pF}_0 = 6.9$ (for soil at the oven-dry state) as a standardized anchor point (i.e., $\theta = 0$, $\mathrm{pF} = 6.9$) when pF_0 is not possible to be measured.

2.5 | Dexter et al. model

Like Durner (1994), Dexter et al. (2008) verified that, for many soils, a unimodal equation could not fit the water retention phenomenon well. Thus, they have proposed the exponential double Equation 5 for bimodal soils (e.g., soils having prominently both textural and structural porosity):

$$w = C + A_1 \left[\exp\left(\frac{-h}{h_1}\right) \right] + A_2 \left[\exp\left(\frac{-h}{h_2}\right) \right] \tag{5}$$

where w is the gravimetric water content (g g^{-1}), C is the residual water content (g g^{-1}), A_1 and A_2 are fitting parameters with gravimetric water content units (e.g., g g^{-1}), and h_1 and h_2 are fitting parameters with matric potential units (e.g., hPa). In Dexter's equation, C is equivalent to the θ_r previously described. Furthermore, Dexter et al. (2008) suggest that the parameters should be fitted in terms of gravimetric water content. To be consistent with the equation originally presented, we describe units and nomenclature of the Equation 5 following Dexter et al. (2008).

3 | MODEL FITTING PROCEDURES

soilphysics makes use of some native implementations in R for parameter estimation purposes. Two statistical methods are used for estimation: nonlinear weighted least squares (NLS), implemented in *fitsoilwater_App*, and maximum likelihood (ML), implemented in *WRC_App*. Maximum likelihood is based on normal distribution for the residual term ε.

As the models are nonlinear, optimization algorithms are used to find the numerical solutions after convergence, namely Gauss-Newton for NLS estimates and Nelder-Mead (1965) for ML estimates.

Three statistics can be used as fitting criteria:

1. Akaike's (1974) information criterion: AIC $= -2\log(L) + 2p$, where L is the maximum of the likelihood function and p is the number of parameters of the model being fitted. This criterion is used in *soilphysics* to select the best-fit model.
2. Mean absolute percentual error: MAPE $= 100/n \sum_{i=1}^{n} |\hat{\varepsilon}_i|/\theta_i$, where $\hat{\varepsilon}_i$ is the ith ($i = 1, 2, \ldots, n$) residual value of the model, associated with the water content θ_i.
3. Pseudo coefficient of determination: $R^2 = 1 - \sum_{i=1}^{n} \hat{\varepsilon}_i^2 / \sum_{i=1}^{n} (\theta_i - \bar{\theta})^2$, where $\bar{\theta}$ is the mean value of water content, as presented by Anderson-Sprecher (1994).

4 | PACKAGE FEATURES AND FUNCTIONALITIES

Three main functionalities can be used to fit the water retention curves presented earlier:

1. The set of functions called *fitsoilwater** (there are five), which consist of GUI (graphical user interface) controls based on package *tcltk* (R Core Team, 2021) for the user to set the initial values of the parameters to be estimated by nonlinear weighted least squares. The output is shown in the console of R. Those functions use as basic input a numeric vector of water content and water tension data, from objects in R. They are available on *soilphysics* version 1.2 or later.
2. The web application *fitsoilwater_App* (Figure 1), built from the *Shiny* (Chang et al., 2021) framework to import data from a spreadsheet (.csv) or text (.txt) file and to fit one of the five models presented before by interacting with the slide controls of the initial values of the parameters to be estimated by nonlinear weighted least squares. The fitting criteria are AIC, MAPE, and pseudo R^2. The app is called from the R console with the command line *fitsoilwater_App()*. It is available from version 4.1 or later.
3. The web application *WRC_App*, built from the Shiny framework in a very similar way to *fitsoilwater_App* but using instead an automatic procedure for estimation of the model parameters, based on maximum likelihood. It allows choosing the models to be fitted and comparing them statistically using AIC, MAPE, and pseudo R^2. If all

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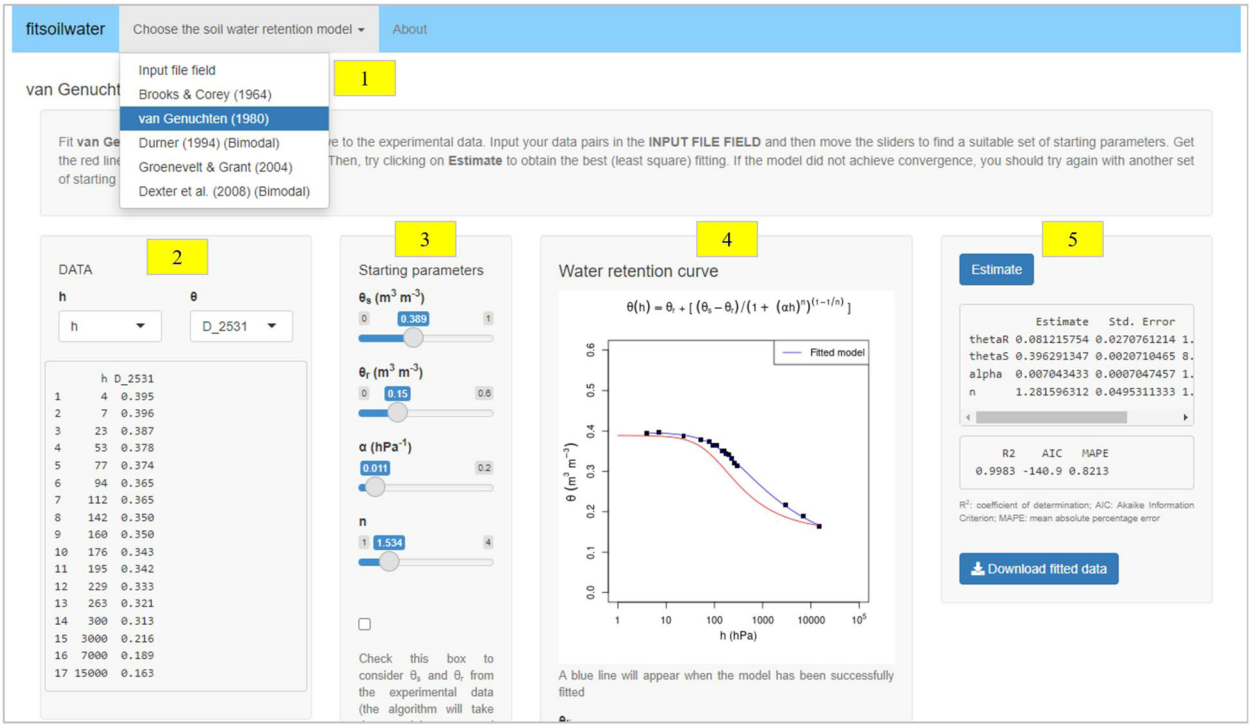

FIGURE 1 Preview of the web application *fitsoilwater_App* of the R package *soilphysics*. [1] Input file field and available water retention models; [2] panel with the inputted dataset; [3] slide controls for initial starting values; [4] interactive plot of the water retention curve; [5] output and statistical panel

models are selected, the one with the lowest AIC will be returned. The app is called from the R console with the command line *WRC_App()*. It is available from version 5.0 or later.

For optimization purposes, the parameters θ_r and θ_s in the respective models are considered as offset in *WRC_App*, meaning that the lowest and highest values, respectively, of the experimental water content data are returned. If needed, statistical estimation of these parameters can be done through *fitsoilwater** and *fitsoilwater_App*.

Both web applications have some basic instructions and an example dataset to be downloaded directly from the app. Both can run online from the main page of the package, at https://arsilva87.github.io/soilphysics.

5 | PERFORMANCE AND COMPARISONS

We compared the performance of the estimates obtained with the *fitsoilwater_App* function with those from RETC software (van Genuchten et al., 1991). For that, we used data from 12 soils available in the UNSONDA 2.0 database (Leij et al., 1996; Nemes et al., 2001). Soils were selected to cover a wide range of soil texture and a full range of matric potentials, from close to saturation up to 15,000 hPa (Table 1).

Table 1 shows that the parameters fitted using RETC and *soilphysics* presented similar values. The greatest differences were found for θ_r for the soils with code 2531 and 2532. Although *fitsoilwater_App* estimated 0.0812 and 0.1003, respectively, RTEC considered both null (i.e., $\theta_r = 0$). On average (all soils), the difference was 0.0152 $m^3\ m^{-3}$ for θ_r. Nonetheless, both curves fitted with *soilphysics* presented higher R^2. Moreover, the *fitsoilwater_App* offers two additional options for fitting θ_r: to "consider θ_r from experimental data," or to "consider $\theta_r = 0$." When the latter was chosen, the set of parameters were the same as those obtained with RETC (data not shown). For all the other three parameters, the correlation between RETC and *soilphysics* estimates was greater than 0.99.

6 | FINAL CONSIDERATIONS

We presented a suite of tools to fit and compare water retention curves that are freely available in the R package *soilphysics*. Due to its user-friendly interface, this package adapts well to didactic and scientific demands, with facilities for estimating parameters of the usual but complex water retention curves. The parameter estimates are usually applicable in numerical simulations of water flux, soil physical quality indicator, and pore size distribution, which are essential for agricultural water management. The software also automatically

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TABLE 1 Comparison of the fitted parameters for the van Genuchten's model using the RETC software and *soilphysics* R package

UNSODA soil data code	Textural class	θ_r		θ_s		α		n		R^2	
		RETC	*soilphysics*	RETC	*soilphysics*	RETC	*soilphysics*	RETC	*soilphysics*	RETC	*soilphysics*
1091	Sandy loam	0.0653	0.0653	0.3055	0.3055	0.0751	0.0751	1.2816	1.2814	0.9947	0.9947
2530	Loam	0.1451	0.1450	0.4346	0.4346	0.0216	0.0217	1.4577	1.4570	0.9822	0.9822
2531[a]	Loam	0.0000	0.0812	0.3981	0.3963	0.0084	0.0070	1.1857	1.2816	0.9978	0.9983
2532[a]	Sandy loam	0.0000	0.1003	0.4281	0.4239	0.0047	0.0039	1.2784	1.5072	0.9966	0.9985
2560	Sandy loam	0.0910	0.0909	0.4822	0.4822	0.1032	0.1033	1.3938	1.3936	0.9975	0.9975
2562	Sand	0.0341	0.0341	0.2650	0.2650	0.0576	0.0576	2.8435	2.8434	0.9993	0.9993
2570	Sandy loam	0.1173	0.1172	0.5626	0.5626	0.0615	0.0616	1.7090	1.7084	0.9922	0.9922
2571	Sandy loam	0.0603	0.0602	0.4819	0.4820	0.0681	0.0681	1.6354	1.6349	0.9948	0.9948
2572	Sand	0.0698	0.0698	0.4764	0.4764	0.0499	0.0500	2.0185	2.0173	0.9929	0.9929
2573	Sand	0.0785	0.0784	0.4412	0.4413	0.0425	0.0425	2.3647	2.3615	0.9860	0.9860
2683	Clay loam	0.1371	0.1370	0.4765	0.4765	0.0722	0.0723	1.3403	1.3399	0.9905	0.9905
2743	Clay loam	0.0862	0.0860	0.7118	0.7118	0.0882	0.0883	1.2131	1.2130	0.9988	0.9988
Mean difference	–	0.0152 ($m^3\ m^{-3}$)		0.0005 ($m^3\ m^{-3}$)		0.0002 (hPa^{-1})		0.0276		–	
Pearson correlation	–	0.6518		0.9999		0.9999		0.9913		–	

Note. UNSODA, Unsaturated Soil Hydraulic Database; θ_r, residual water content; θ_s, saturated water content; α and n, fitting parameters.
[a]The parameters of RETC and *soilphysics* were exactly the same for these soils when the option "consider $\theta_r = 0$" (available in *soilphysics fitsoilwater_App* interface) was assigned.

fits several models, allowing for statistical comparisons to select the best-fit model. Future developments and updates of the R package *soilphysics* should move towards the inclusion of further water retention models and soil water indicators and related functionalities, including recent and traditional relevant models.

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AUTHOR CONTRIBUTIONS

Renato P. de Lima: Conceptualization; Methodology; Software; Writing – original draft; Writing – review & editing. Anderson R. da Silva: Conceptualization; Methodology; Software; Writing – original draft; Writing – review & editing.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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